

LUCIDMSK CME PROGRAM

MODULE 08 — POST-TEST

Pelvis & Hip CT: Fracture Classification Systems

Pelvis · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Young-Burgess LC-II pelvic fracture defining feature:

- A Ipsilateral sacral compression fracture only
- B Ipsilateral sacral fracture + ipsilateral iliac crescent fracture
- C Bilateral anterior + posterior ring disruption
- D Complete SI joint disruption

2 Garden Stage III femoral neck fracture. Defining feature:

- A Incomplete; valgus impaction
- B Complete; NO displacement; trabeculae aligned
- C Complete; PARTIAL displacement; trabeculae of neck NOT aligned with acetabulum
- D Complete; full displacement; head freely rotated

3 Denis Zone II sacral fracture passes through:

- A Lateral to the sacral foramina (sacral ala)
- B The sacral foramina themselves
- C The spinal canal
- D The S1 endplate only

4 Pipkin Type II femoral head fracture:

- A Fragment inferior to fovea (non-weight-bearing)
- B Fragment SUPERIOR to fovea (weight-bearing articular surface)
- C Head fracture + femoral neck fracture
- D Head fracture + acetabular fracture

5 Pauwels Type III femoral neck fracture:

- A Horizontal (<30°); compressive; stable
- B Intermediate (30–50°); mixed
- C Vertical (>50°); SHEAR forces; UNSTABLE; highest AVN risk
- D Impacted valgus; most stable

6 Schatzker Type VI tibial plateau fracture:

- A Lateral plateau split only
- B Lateral plateau split + depression
- C Bicondylar without metaphyseal dissociation
- D Bicondylar + metaphyseal-diaphyseal DISSOCIATION

7 Judet-Letournel "T-shaped" acetabular fracture involves:

- A Anterior column only
- B Posterior column + vertical stem through obturator foramen
- C Both columns; posterior wall associated
- D Anterior column + posterior hemitransverse

8 Young-Burgess APC-III pelvic fracture. Expected injury:

- A Intact posterior SI ligaments
- B Partial SI disruption; posterior SI intact
- C Complete SI disruption; both anterior AND posterior SI ligaments torn
- D Contralateral open-book only

9 Both-column acetabular fracture pathognomonic CT finding:

- A Intact dome with anterior column fracture

- | | |
|----------|--|
| B | Both column Fx; at least one segment attached to axial skeleton |
| C | Both column Fx; NO segment attached to axial skeleton; "spur sign" |
| D | Posterior wall + transverse only |

10 Posterior acetabular wall fracture involving >50% of the posterior wall after hip dislocation:

- | | |
|----------|--|
| A | Stable; conservative management |
| B | Borderline; CT stress exam |
| C | Posterior hip INSTABILITY — surgical fixation of posterior wall required |
| D | No change from smaller fragments |

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	C	B	B	C	D	B	C	C	C

Q1 Young-Burgess LC-II pelvic fracture defining feature:

Answer: B

A Ipsilateral sacral compression fracture only

✓ **Ipsilateral sacral fracture + ipsilateral iliac crescent fracture**

C Bilateral anterior + posterior ring disruption

D Complete SI joint disruption

EXPLANATION YB LC-II = ipsilateral sacral buckle fracture + ipsilateral posterior iliac crescent (iliac wing) fracture. LC-I = sacral compression alone. LC-III = LC-II + contralateral open-book.

Q2 Garden Stage III femoral neck fracture. Defining feature:

Answer: C

A Incomplete; valgus impaction

B Complete; NO displacement; trabeculae aligned

✓ **Complete; PARTIAL displacement; trabeculae of neck NOT aligned with acetabulum**

D Complete; full displacement; head freely rotated

EXPLANATION Garden III = complete fracture with PARTIAL displacement — femoral head displaced but trabeculae of neck no longer align with acetabular trabeculae. Garden IV = complete displacement with head freely rotated.

Q3 Denis Zone II sacral fracture passes through:

Answer: B

A Lateral to the sacral foramina (sacral ala)

✓ **The sacral foramina themselves**

C The spinal canal

D The S1 endplate only

EXPLANATION Denis Zone II = transforaminal (through the sacral foramina). Zone I = lateral to foramina (ala). Zone III = medial (spinal canal). Zone II nerve injury rate ~28%. Zone III highest neurological risk.

Q4 Pipkin Type II femoral head fracture:

Answer: B

A Fragment inferior to fovea (non-weight-bearing)

✓ **Fragment SUPERIOR to fovea (weight-bearing articular surface)**

C Head fracture + femoral neck fracture

D Head fracture + acetabular fracture

EXPLANATION Pipkin II = fragment superior to fovea capitis = weight-bearing dome involved. Type I = inferior to fovea (non-WB; better prognosis). Type III = head + femoral neck. Type IV = head + acetabular fracture.

Q5 Pauwels Type III femoral neck fracture:

Answer: C

A Horizontal ($<30^\circ$); compressive; stable

B Intermediate ($30-50^\circ$); mixed

✓ **Vertical ($>50^\circ$); SHEAR forces; UNSTABLE; highest AVN risk**

D Impacted valgus; most stable

EXPLANATION Pauwels III = most vertical fracture line ($>50^\circ$) — high shear stress, highest risk of non-union and AVN. Pauwels I = most horizontal ($<30^\circ$), compressive forces, most stable.

Q6 Schatzker Type VI tibial plateau fracture:

Answer: D

A Lateral plateau split only

B Lateral plateau split + depression

C Bicondylar without metaphyseal dissociation

✓ **Bicondylar + metaphyseal-diaphyseal DISSOCIATION**

EXPLANATION Schatzker VI = bicondylar WITH metaphyseal-diaphyseal dissociation (most severe). Type V = bicondylar without dissociation. Types I–III = lateral plateau only. Type IV = medial plateau.

Q7 Judet-Letournel "T-shaped" acetabular fracture involves:

Answer: B

A Anterior column only

✓ **Posterior column + vertical stem through obturator foramen**

C Both columns; posterior wall associated

D Anterior column + posterior hemitransverse

EXPLANATION T-shaped = transverse fracture + vertical stem through the obturator foramen/ischiopubic region. Both-column fracture = no segment attached to axial skeleton (floating acetabulum — "spur sign").

Q8 Young-Burgess APC-III pelvic fracture. Expected injury:

Answer: C

A Intact posterior SI ligaments

B Partial SI disruption; posterior SI intact

✓ **Complete SI disruption; both anterior AND posterior SI ligaments torn**

D Contralateral open-book only

EXPLANATION APC-III = complete SI joint disruption — both anterior AND posterior SI ligaments torn = maximum pelvic instability. Associated with major internal iliac vessel hemorrhage.

Q9 Both-column acetabular fracture pathognomonic CT finding:**Answer: C**

- A Intact dome with anterior column fracture
- B Both column Fx; at least one segment attached to axial skeleton
- ✓ **Both column Fx; NO segment attached to axial skeleton; "spur sign"**
- D Posterior wall + transverse only

EXPLANATION Both-column fracture = ALL articular segments detached from the axial skeleton. The pathognomonic "spur sign" = isolated fragment of iliac wing posterior to the fracture. Most common complex acetabular pattern.

Q10 Posterior acetabular wall fracture involving >50% of the posterior wall after hip dislocation:**Answer: C**

- A Stable; conservative management
- B Borderline; CT stress exam
- ✓ **Posterior hip INSTABILITY — surgical fixation of posterior wall required**
- D No change from smaller fragments

EXPLANATION >50% posterior wall fragment = posterior hip instability after reduction. Fragments >25–40% may require fixation depending on CT stress exam. Large posterior wall fractures require ORIF to restore stability.